

# MD. NOMAN BISWAS SIBLY

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🌐 Portfolio Website —  LinkedIn —  GitHub —  Scholar

📍 Shahid President Ziaur Rahman Hall, RUET, Kazla, Rajshahi-6204, Bangladesh

## EDUCATION

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**Rajshahi University of Engineering & Technology (RUET)**

*Dec 2022 – Present*

*B.Sc. in Electrical & Computer Engineering*

CGPA: 3.56/4.00

(after 6th semester)

**BAF Shaheen College, Jashore**

*Jul 2019 – Dec 2021*

*HSC – Science*

GPA: 5.00/5.00

## RESEARCH

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### Publications

- First Author, “Which Reliability Signal to Trust? Output vs. Representation Space Uncertainty Under Distribution Shift in Pathology Foundation Models”, accepted at *MI4MedFM Workshop, MICCAI 2026*. [\[Code\]](#)
- Second Author, “A Comprehensive Review of Convolutional Neural Networks: Foundations, Enhancements, and Applications”, *Neural Computing and Applications* (Springer, Q1, IF 5.102), 38(4):56, 2026. [\[DOI\]](#)
- First Author, “Noise-Robust Gallbladder Ultrasound Image Classification Using U-Net GAN Denoising and Vision Transformer”, *IEEE QPAIN 2026*. [\[DOI\]](#)

### Manuscript in Preparation

- First Author, “JTADC-Net: An Adversarial Joint Task-Aware Denoising and Classification Network for Noise Robust Gallbladder Disease Diagnosis” — joint adversarial denoising and classification for noise-robust gallbladder diagnosis. [\[Code\]](#)

### Ongoing Research

- Trustworthy AI in Healthcare (Thesis)

## EXPERIENCE

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**Research Assistant (Part-time, Remote)**

2026 – Present

*ELITE Research Lab*

*Bangladesh*

- Working on mechanistic interpretability of neural networks to make predictions explainable.

**AI & ML Trainee**

2025 – 2026

*BDSET*

*Bangladesh*

- Applied ML training: preprocessing, model building, tuning, evaluation (Python, TensorFlow, Scikit-learn).

**Programming Mentor**

2024 – 2025

*RUET Analytical Programming Lab*

*Rajshahi*

- Taught algorithms/debugging and guided students in competitive programming and contest preparation.

## PROJECTS

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- ↪ **JTADC-Net: Noise-Robust Gallbladder Disease Classification:** Developed a GAN-assisted ViT framework for robust gallbladder ultrasound classification under noisy conditions, packaged as a real-time medical image classification web app with deployment support. [\[Code\]](#) [\[Demo\]](#) [2025]
- ↪ **TruthLayer-Gemma4: Trustworthy Vision-Language AI:** Explored reliability, hallucination reduction, and verification strategies in multimodal vision-language systems, investigating trustworthy AI methods for safer and more interpretable inference. [\[Code\]](#) [2025]
- ↪ **ATmega Neural Network – Digit Recognizer:** Built a lightweight neural network for digit classification on ATmega microcontrollers using optimized integer computations, with memory-efficient TinyML inference pipelines for constrained embedded environments. [\[Code\]](#) [2025]

## PROBLEM SOLVING

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[Codeforces](#) (Pupil, max 1263; 1186 solved) · [LeetCode](#) (125 solved) · [VJudge](#) (378 solved)

## CERTIFICATES & ACHIEVEMENTS

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|-----------------------------------------------------------------------------------------------------------|-----------------------|
| <b>6th Place – AI Hackathon, DUET</b><br><i>Dept. of ECE, RUET</i>                                        | 2026<br><i>Verify</i> |
| · Placed 6th at the inter-university AI hackathon hosted at DUET                                          |                       |
| <b>AI for Immersive Technology – 288-Hour Training</b><br><i>BDSET, Bangladesh Hi-Tech Park Authority</i> | 2026<br><i>Verify</i> |
| · Six-month program in AI, machine learning, deep learning, and immersive technologies (VR, AR, MR, XR)   |                       |
| <b>Machine Learning Specialization (Andrew Ng)</b><br><i>DeepLearning.AI &amp; Stanford</i>               | 2024<br><i>Verify</i> |
| · Completed 3-course ML specialization (Supervised ML, Advanced ML, ML Engineering)                       |                       |
| <b>CS50: Databases with SQL</b><br><i>Harvard University</i>                                              | 2024<br><i>Verify</i> |
| · Covered SQL, schemas, indexing, transactions, triggers, and database design                             |                       |
| <b>CS50: Introduction to Computer Science</b><br><i>Harvard University</i>                                | 2023<br><i>Verify</i> |
| · Learned Python, algorithms, data structures, and problem-solving                                        |                       |
| <b>2nd Runner-Up – ECE Youth Programming Contest</b><br><i>RUET</i>                                       | 2023                  |
| · Ranked 3rd in the department-level competitive programming contest                                      |                       |

## TECHNICAL SKILLS

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- ↪ **Languages:** C, C++, Python, JavaScript, SQL
- ↪ **Data Science & ML Tools:** TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib
- ↪ **Web Development:** HTML, CSS, Django, MySQL
- ↪ **Other Tools:** Git, GitHub, Linux, Jupyter Notebook, Data Analysis